

Introduction to Stochastic Processes (STAT50017.01)

Answer ALL SIX questions. Marks are shown in square brackets.

Question 1

Let X and Y be independent exponential random variables with common rate $\lambda = 1$. Define

$$U = \frac{X}{X+Y} \quad \text{and} \quad V = X + Y.$$

- (α) Find the joint density of (U, V) . [5 marks]
- (α) Determine whether U and V are independent, and identify their marginal distributions. [4 marks]
- (α) Find $\mathbb{P}(X < 2Y \mid X + Y < 1)$. [4 marks]

Question 2

Customers arrive at a service desk according to a Poisson process with rate $\lambda = 4$ per hour.

- (α) Find the probability that exactly three customers arrive in the first 30 minutes and exactly five customers arrive in the first hour. [5 marks]
- (α) Given that exactly five customers arrive in the first hour, find the probability that exactly three of them arrive in the first 30 minutes. [5 marks]
- (α) Given that exactly five customers arrive in the first hour, find the expected time of the third arrival. [5 marks]

Question 3

Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ . Given $N(T) = n$, denote the arrival times by $0 < S_1 < \cdots < S_n < T$. Define $M = \max\{S_i, T - S_n\}$. Find

$$\mathbb{P}(M \leq x \mid N(T) = n), \quad 0 < x < T.$$

[18 marks]

Question 4

Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ . Fix a deterministic time $t > 0$. Let

$$A_t = t - S_{N(t)} \quad \text{and} \quad B_t = S_{N(t)+1} - t,$$

with the convention that $A_t = t$ if $N(t) = 0$.

(α) Find the distribution of B_t . [7 marks]

(α) Find the distribution of A_t . [7 marks]

Question 5

Suppose that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are independent Poisson processes with rates λ_1 and λ_2 respectively. Let $N(t) = N_1(t) + N_2(t)$, and T_n be the time of the n th arrival of the process $\{N(t)\}$. Define

$$G_n = T_n - \max\{S_j^{(1)} : S_j^{(1)} < T_n\},$$

where $S_j^{(1)}$ is the j th arrival time of the process $\{N_1(t)\}$. If there is no arrival from process $\{N_1(t)\}$ before T_n , define $G_n = T_n$. Also define

$$H_n = \min\{S_j^{(2)} : S_j^{(2)} > T_n\} - T_n,$$

that is, H_n is the waiting time from T_n until the next arrival from the process $\{N_2(t)\}$.

(α) Given $T_n = t$, find the conditional distribution of G_n . [8 marks]

(α) Compute $\mathbb{E}[G_n \mid T_n = t]$. [6 marks]

(α) Find the distribution of H_n . [6 marks]

Question 6

Let $\{X(t), t \geq 0\}$ be a compound Poisson process

$$X(t) = \sum_{i=1}^{N(t)} Y_i,$$

where $\{N(t), t \geq 0\}$ is a Poisson process with rate λ , and $Y_1, Y_2, \dots$ are i.i.d. exponential random variables with rate μ independent of $N(t)$.

(α) Find the moment generating function of $X(t)$. [6 marks]

(α) Find $\mathbb{P}(X(t) = 0)$. For $x > 0$, express the density function $f_{X(t)}(x)$ as an infinite series. [7 marks]

(α) For fixed $a > 0$, find $\mathbb{E}[e^{-aX(t)} \mid N(t) \geq 1]$. [7 marks]